

MGT 595: Quantitative Investing

Problem Set 1 — Part 1

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0 Dataset Construction

0.1 Investment Philosophy

What investment philosophy guided the construction of the universe?

This portfolio was constructed with a focus on “Quality Growth.” Because of our shared interest in Information Technology and Health Care, we assigned greater weights to these sectors for their long-term growth potential and profitability. We also allocated a meaningful share to Energy, where structural demand and different economic drivers could help balance the growth-oriented holdings. Meanwhile, the portfolio maintains exposure to all 11 GICS sectors to reduce concentration risk and avoid relying too heavily on the performance of only a few industries.

0.2 Candidate Pool and Screening Method

How were the stocks screened and selected?

Once the number of stocks required from each sector had been determined, we formed a candidate pool approximately twice the size of the eventual allocation. The candidate pools included both large, established companies and smaller firms with market capitalizations below \$5 billion.

We then compared the candidates using a simplified multifactor framework that considered profitability, revenue growth, valuation, leverage, price momentum, and market risk. Because the portfolio follows a “Quality Growth” philosophy, greater emphasis was placed on growth and profitability. Valuation, debt, Beta, and recent price movements were also considered to keep the overall risk at a reasonable level.

0.3 Final Stock Universe

Does the final universe satisfy the assignment requirements?

The final universe contains exactly 50 U.S.-listed stocks and maintains exposure across all 11 GICS sectors. It includes five companies with market capitalizations below \$5 billion: OII, ADUS, LNN, PLUS, and PLAB. Therefore, the universe satisfies both the sector-diversification and small-cap requirements of the assignment.

After the stocks had been selected, the 50 tickers were alphabetized before constructing the nested portfolios used in Part 1. The first 5, 10, and 25 stocks were therefore determined by this fixed alphabetical order rather than by their screening rankings. This makes the comparison across portfolio sizes more neutral by preventing us from intentionally placing our most preferred stocks in the smaller portfolios.

0.4 Data Collection

How were the historical prices obtained?

Adjusted monthly closing prices for all 50 stocks and the S&P 500 (^GSPC) were downloaded using `yfinance` with `interval="1mo"` and `auto_adjust=True`. The final sample covers January 2010 through the most recent complete month and contains 200 monthly return observations.

0.5 Missing-Data Validation

Were any stocks or months removed because of missing data?

All 50 stocks had complete monthly coverage over the common sample period. Therefore, no ticker exceeded the assignment's 5% missing-data threshold, no ticker was dropped or replaced, and no month was removed during complete-case alignment.

0.6 Extreme-Return Validation

Were potentially unusual monthly returns investigated?

As part of the data-validation process, monthly returns below -50% or above $+200\%$ were flagged for investigation rather than automatically removed. Three observations met this criterion: EXEL returned approximately -63.0% in September 2014, while OII and OKE each declined by approximately 70% in March 2020.

The latter two observations coincided with the COVID-19 market disruption and the collapse in oil prices. Because the flagged observations appeared to represent genuine market movements rather than data errors, they were retained in the final dataset.

0.7 Bias Acknowledgement

How should selection and survivorship bias affect the interpretation of the results?

Our sample is subject to both selection and survivorship bias. The selection process intentionally favored companies with stronger current fundamentals, particularly in Information Technology and Health Care. As a result, the portfolio's returns, volatility, and covariance structure may not be representative of the broader U.S. stock market.

In addition, requiring continuous historical data back to January 2010 restricted the sample to companies that are still listed today. Firms that failed, were delisted, or were acquired before the selection date were excluded by construction. Therefore, average returns may appear higher and downside risk may appear lower than they would in a universe that included unsuccessful firms, although this effect is not guaranteed for every excluded company.

The diversification patterns reported in Part 1 should consequently be interpreted as evidence for this particular group of selected surviving firms, rather than as general estimates for all stocks that were investable in January 2010. Alphabetizing the tickers makes the comparison among the nested portfolios more neutral, but it does not eliminate the selection bias in the original 50-stock universe.

0.8 Dataset Summary

- **Investment philosophy:** Quality Growth
- **Number of stocks:** 50
- **GICS sectors represented:** 11

- **Stocks below \$5 billion:** 5
- **Market benchmark:** S&P 500 ($\sim$ GSPC)
- **Return frequency:** Monthly
- **Final number of monthly returns:** 200
- **Stocks dropped or replaced:** None
- **Months removed during alignment:** None
- **Extreme observations flagged and retained:** 3

Question 1(a): Diversification and Basic Statistics

We form equal-weight portfolios using the first 5, 10, 25, and all 50 tickers in fixed alphabetical order. For each portfolio, the sample mean and sample standard deviation of monthly returns were computed.

Table 1: Equal-weight portfolio statistics

N	Sample Mean	Sample Std. Dev.
5	1.6362%	4.9518%
10	1.6393%	4.4471%
25	1.5201%	4.5803%
50	1.5960%	4.4970%

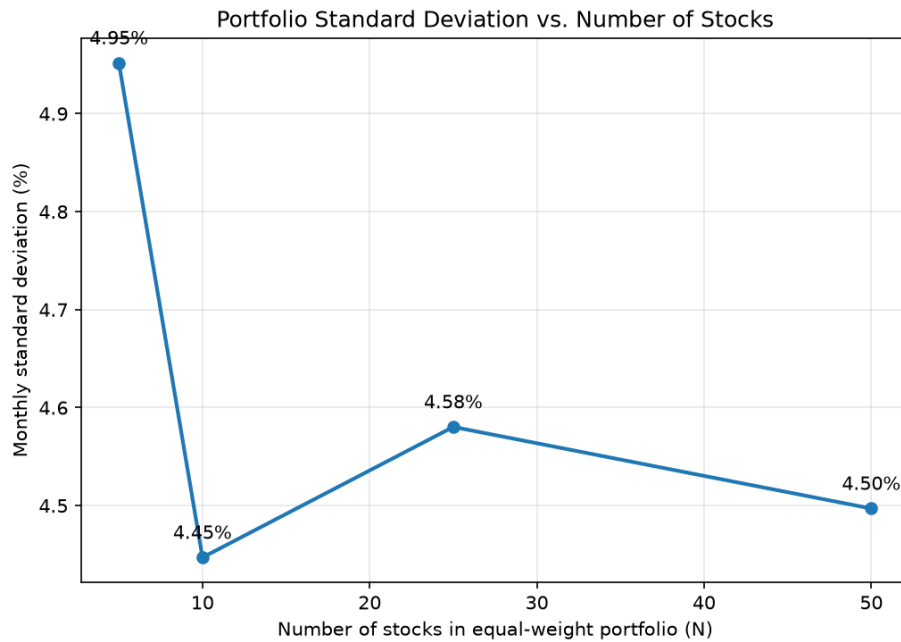


Figure 1: Portfolio standard deviation vs. number of stocks (N)

Discussion. Although the curve is not perfectly monotonic, it shows an overall downward and flattening pattern. This is broadly consistent with diversification theory: adding stocks generally reduces firm-specific risk, but a realized nested sample does not guarantee that volatility falls with every addition.

Despite the decline, the standard deviation would not reach zero even if the portfolio grows larger. As the portfolio becomes larger, firm-specific risk declines, but common covariance and systematic market risk remain and cannot be diversified away.

Question 1(b): Variance Decomposition

For each equal-weight portfolio, we decomposed total variance into a contribution from individual stock variances and a contribution from covariances, using

$$\text{Var}(r_p) = \frac{1}{N} \overline{\text{Var}} + \frac{N-1}{N} \overline{\text{Cov}},$$

where $\overline{\text{Cov}}$ is backed out algebraically from the portfolio variance and the average individual variance, without directly estimating the full pairwise covariance matrix.

Table 2: Variance decomposition by portfolio size

N	Avg. Var.	Avg. Cov.	Port. Var.	Var. Contrib.	Cov. Contrib.	% from Var.
5	0.007284	0.001244	0.002452	0.001457	0.000995	59.41%
10	0.006481	0.001477	0.001978	0.000648	0.001330	32.77%
25	0.007113	0.001889	0.002098	0.000285	0.001813	13.56%
50	0.007485	0.001911	0.002022	0.000150	0.001873	7.40%

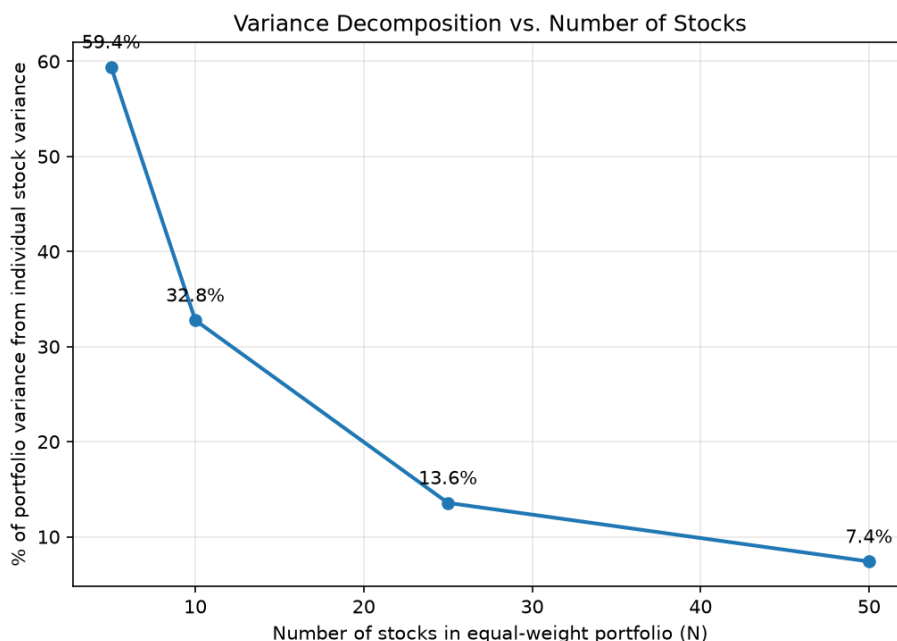


Figure 2: Percentage of portfolio variance from individual stock variance vs. N

Discussion. Compared with Question 1(a), this curve shows a much smoother decline. As the number of stocks increases, the percentage of portfolio variance attributable to individual stock variances decreases, making the covariance contribution increasingly important.

This is consistent with diversification theory. Because the individual-variance component is divided by N^2 , it is gradually diluted as the portfolio grows and may approach zero. However, total portfolio variance does not approach zero because covariance persists as stocks remain exposed to shared market and economic conditions.